

What we're doing

We fold 96 natural eval-val and 19 designed eval-denovo proteins with Helico. For each target we give Helico 0, 10, 20, ... contacts up to exactly L, ranked by the latest MarFold model (exp277 step 266344). Each cut has three diffusion samples. The full grid has 3,089 cuts and 9,267 predictions.

Why

The usual top-L cut is only one operating point. Sweeping the number of contacts reveals the best possible structure available in the grid and tests whether Helico's own confidence head selects it. We report GDT-TS, IDDT, TM-score and RMSD, separately for natural and designed proteins.

Results so far

All 9,267 structures completed for 96 eval-val and 19 eval-denovo targets. On eval-val, GDT-TS is 0.6242 oracle / 0.5277 Helico-ranked / 0.5015 top-L; the ranked sweep beats top-L by +0.0262 [0.0039, 0.0508]. On eval-denovo the same values are 0.9341 / 0.8990 / 0.8503. Confidence chooses zero contacts for 8/19 designs, showing that top-L is the wrong universal policy.

What the gap means

The sweep already generates substantially better natural-protein structures: the oracle is +0.0965 GDT-TS above the confidence choice on eval-val. Helico's confidence model recovers some of the opportunity but remains the bottleneck. On designs the oracle gap is smaller (+0.0352), and confidence usually prefers far fewer contacts (median 0.056L) than it does on natural proteins (0.631L).

Comparison with structure predictors

On the identical eval-val cohort, GDT-TS is 0.8583 for Protenix-v2 + MSA, 0.8000 for ESMFold2, 0.7376 for ESMFold, 0.6242 for the Helico sweep oracle, 0.5277 for confidence-selected Helico, and 0.5015 for top-L Helico. On eval-denovo, ESMFold2 and the Helico oracle both reach about 0.934; confidence-selected Helico reaches 0.8990 versus 0.8503 for top-L. The oracle uses ground truth and measures headroom rather than a deployable policy.

Low-MSA-depth proteins

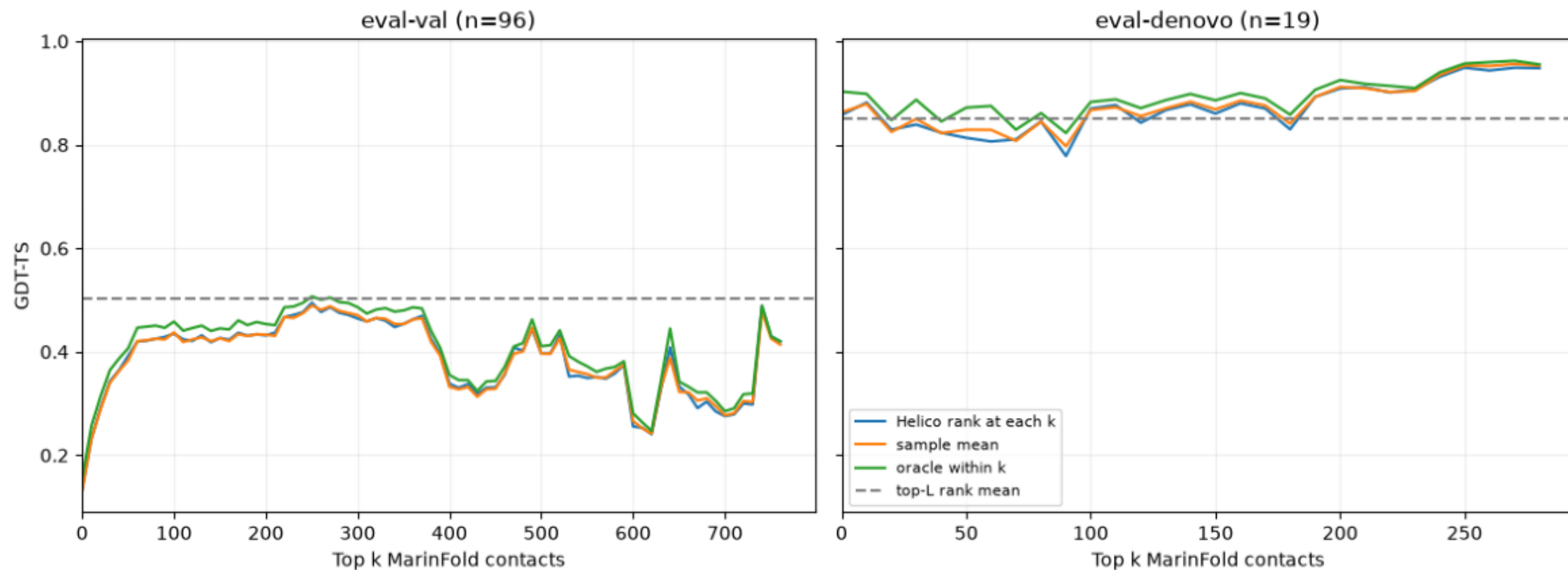
No exp311 eval-val protein has MSA depth ≤ 10 ; five have depth ≤ 100 . On those five, GDT-TS is 0.6223 for Protenix-v2 + MSA, 0.5014 for ESMFold2, 0.4394 for the Helico oracle, 0.3745 for confidence-selected Helico, and 0.2676 for top-L. Among 15 eval-denovo designs at depth ≤ 10 , confidence-selected Helico reaches 0.8977 versus 0.8389 for top-L, 0.8949 for Protenix-v2 + MSA, and 0.9330 for both ESMFold2 and the Helico oracle. These inclusive cuts use the exact A3M given to Protenix, query included.

Conclusion

Search the contact count and use Helico confidence instead of always folding top-L. This improves both natural and designed accuracy. The next useful gain is better confidence ranking: the structures needed for a much higher eval-val score are already present in the sweep.

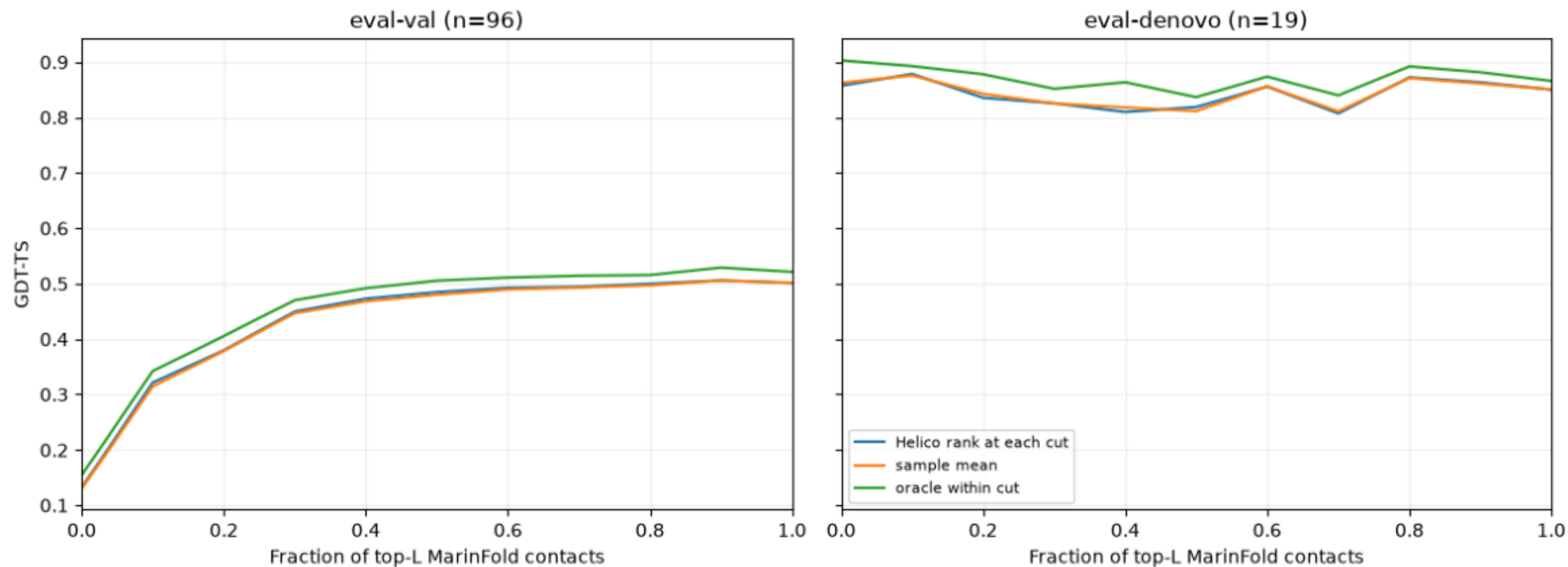
gdt_ts_by_contact_count.png

Unpaired mean at each absolute k; targets with $L < k$ leave the cohort. Dashed line is the all-target top-L mean.



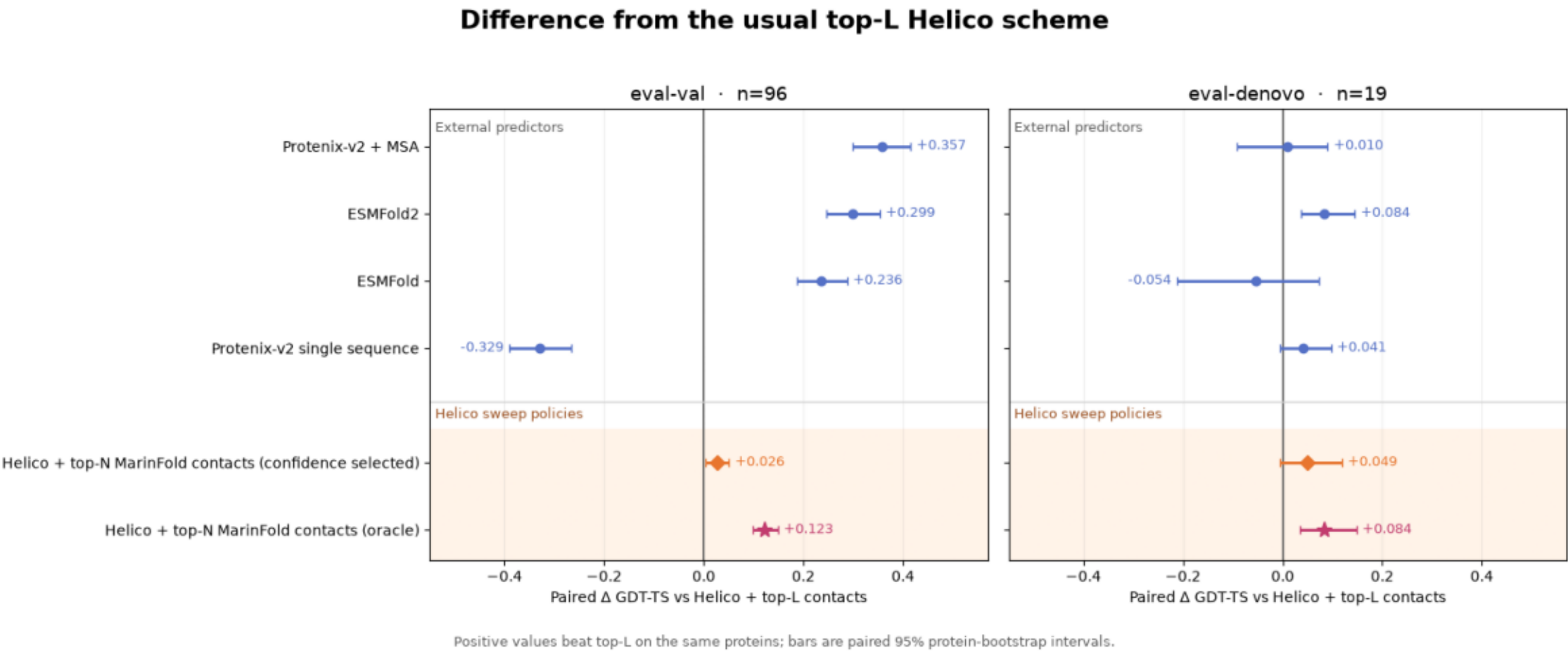
gdt_ts_by_fraction_L.png

Paired target cohort at every fraction of L; each point uses the nearest actual 10-contact cut.



gdt_ts_delta_vs_top_l.png

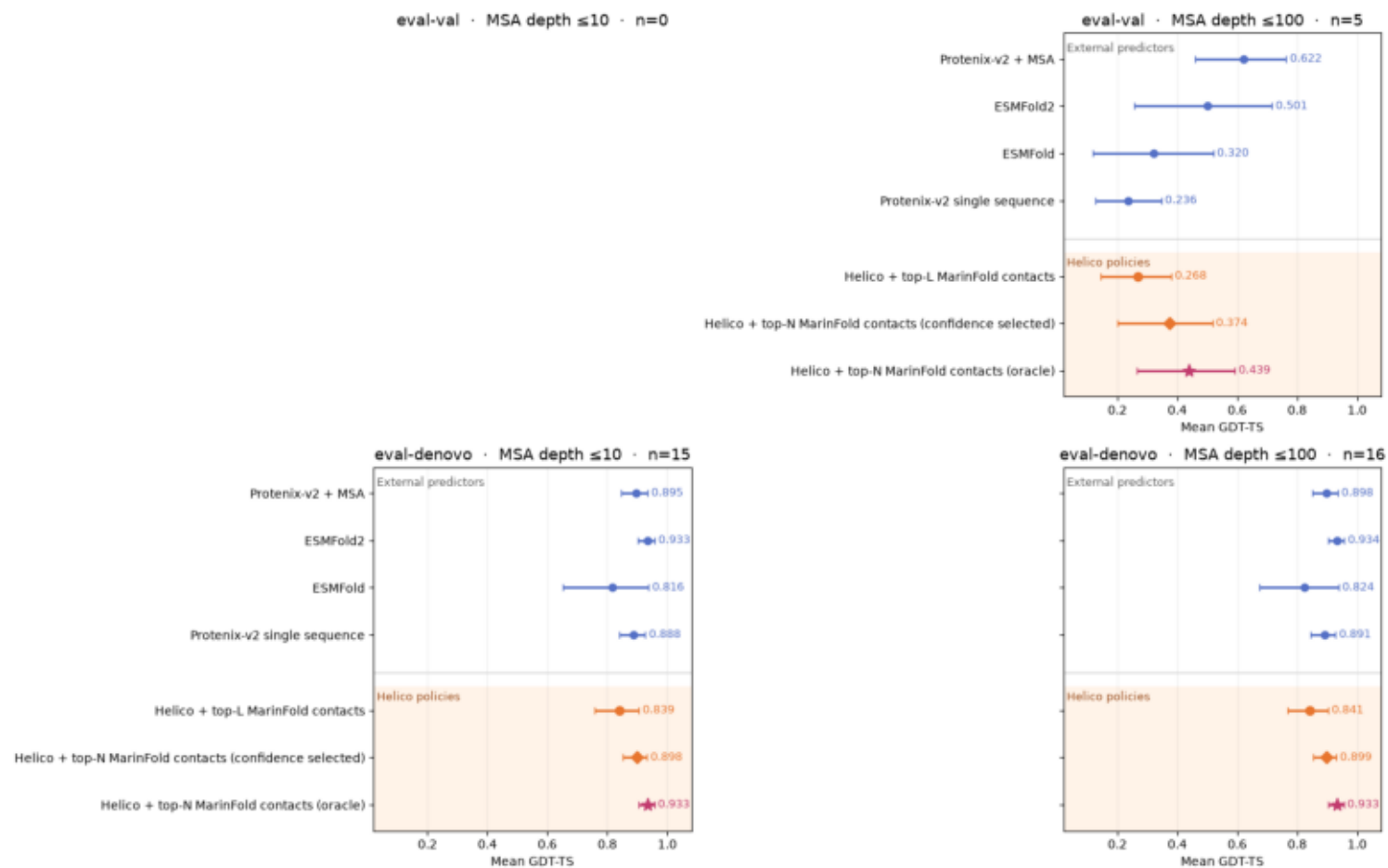
Paired mean GDT-TS difference from Helico with exact top-L Marifold contacts. Positive values are better; intervals resample matched proteins.



gdt_ts_low_msa_predictor_comparison.png

GDT-TS at inclusive MSA-depth thresholds of 10 and 100 sequences. Depth is measured on the exact A3M supplied to Protenix-v2 + MSA; thresholds are cumulative and all methods use paired targets. Points show actual mean GDT-TS values rather than differences from top-L Helico.

Structure accuracy at low MSA depth



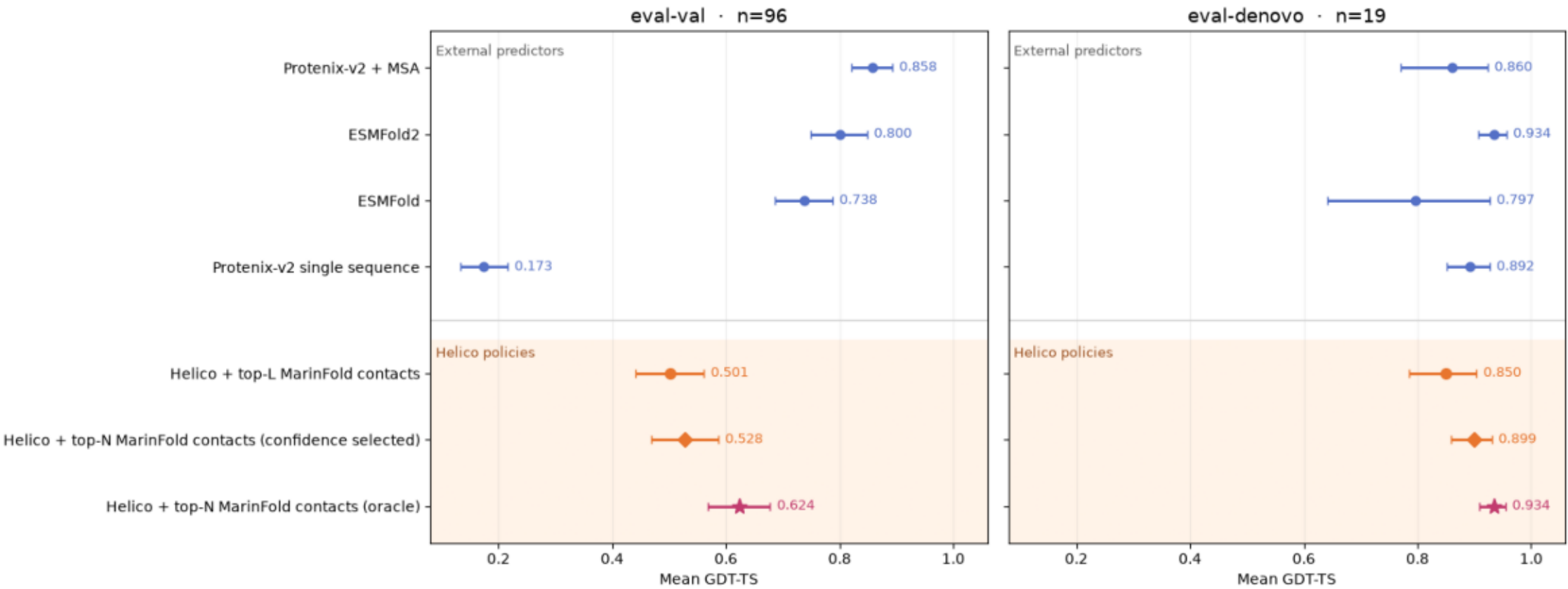
Inclusive, cumulative depth cuts. Points are target means; bars are 95% protein-bootstrap intervals. The oracle uses ground truth.

\$ plot_low_msa_comparison.py

gdt_ts_predictor_comparison.png

Target-mean GDT-TS for four external structure predictors and three exp311 Helico policies. All methods use the same 96 eval-val and 19 eval-denovo targets; intervals resample proteins.

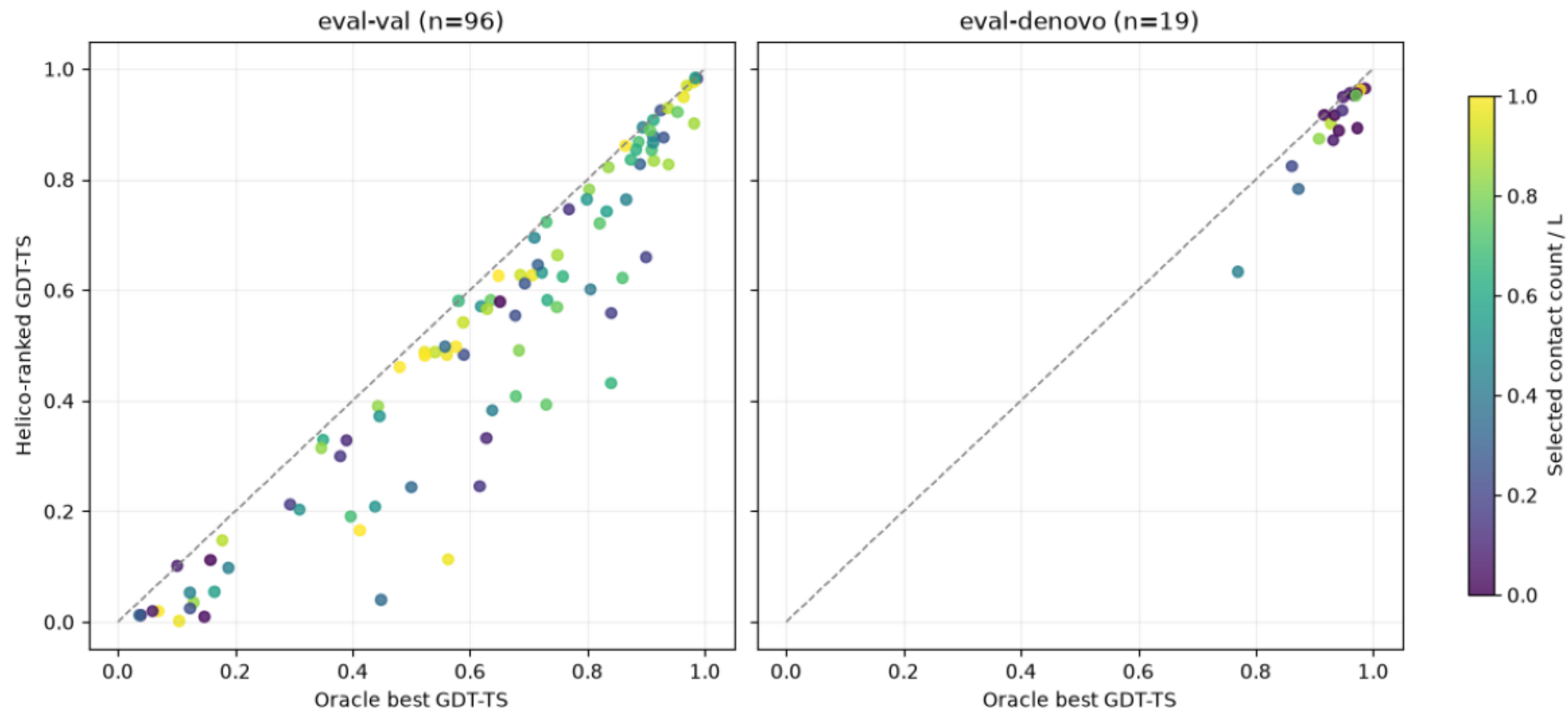
Structure accuracy on identical target cohorts



Points are target means; bars are 95% protein-bootstrap intervals. The oracle uses ground truth to choose across contact counts and diffusion samples.

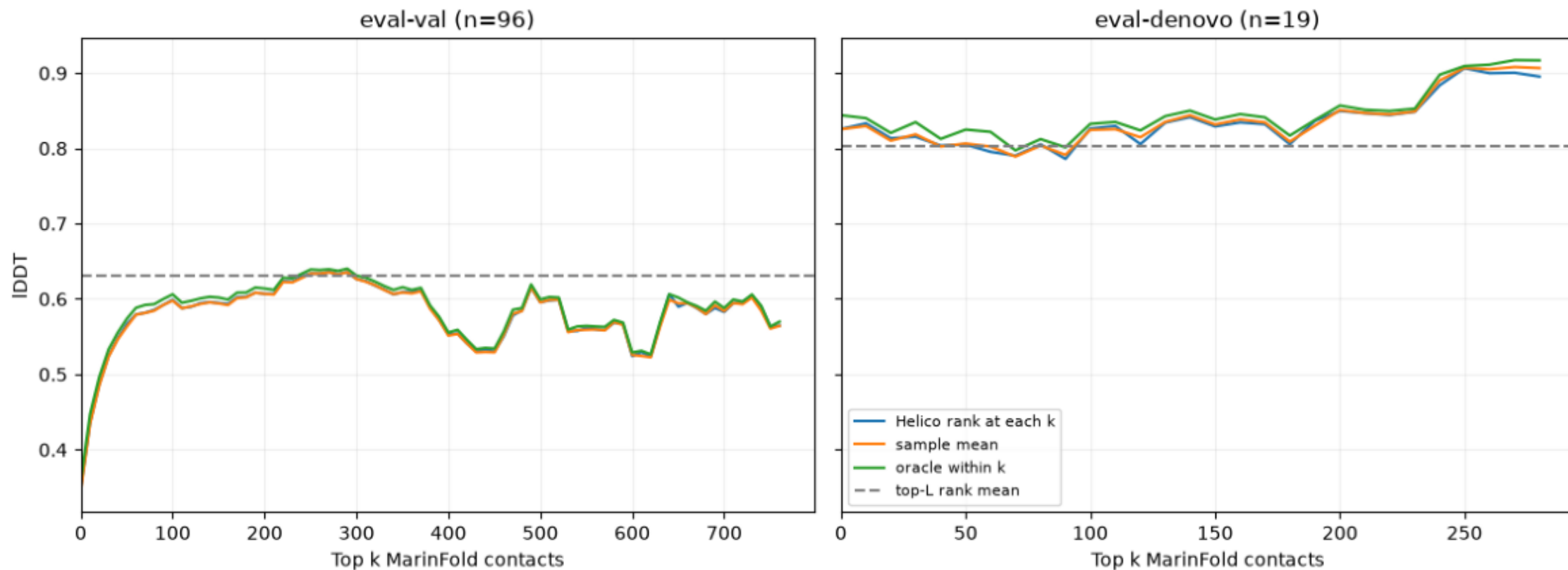
gdt_ts_selection_gap.png

Each protein's best GDT-TS in the full sweep versus the structure Helico confidence ranks first.



Iddt_by_contact_count.png

Unpaired mean at each absolute k; targets with $L < k$ leave the cohort. Dashed line is the all-target top-L mean.



Iddt_by_fraction_L.png

Paired target cohort at every fraction of L; each point uses the nearest actual 10-contact cut.

